BRIGHTLIGHT TUTORIALS
DATA ANALYTICS
Exercise 4

1 SELECT
e. Employee ID, e. First Name, e. Last Name, e. Department,
e. Salary,
p. Project ID, p. Project Name, p. Budget, p. Status
FROM Employees e
INNER JOIN Projects p
ON e. Employee ID = p. Employee ID;

Emp ID	Name	Dept	Salary	Project	Status
1	John	IT	70 000	AI Development	Completed
1	John	IT	70000	Cybersecurity Audit	Pending
2	Alice	HR	60000	Employee training	Ongoing
3	Bob	Finance	75000	Financial Analysis	Ongoing
5	Emma	Sales	65000	Market expansion	Completed
6	Michael	Finance	80000	Risk management	Pending

2 SELECT
e. Employee ID, e. First Name, e. Last Name, e. Department,
e. salary
p. Project ID, p. Project Name, p. Budget, p. Status
FROM Employees e
LEFT JOIN projects p
ON e. Employee ID = p. Employee ID;

Employees without projects
David

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4 SELECT
e. Employee ID, e. First Name, e. Last Name,
e. Department, e. Salary,
p. Project ID, p. Project Name, p. Budget, p. Status
FROM Employees e
FULL OUTER JOIN projects p
ON e. Employee ID = p. Employee ID;

EmpID	Name	Dep	Salary	Project	Status
1	John	IT	70000	AI Development	Completed
1	John	IT	70000	Cybersecurity Audit	Pending
2	Alice	HR	60000	Employee training	Ongoing
3	Bob	Finance	75000	Financial Analysis	Ongoing
4	David	IT	72000	NULL	NULL
5	Emma	Sales	65000	Market Expansion	Completed
6	Michael	Finance	80000	Risk Management	Pending

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7 SELECT Employee ID, First Name, Last Name, Department, Salary
FROM Employees
WHERE Salary > 70000 ;

Employee ID	Name	Department	Salary
3	Bob	Finance	75000
4	David	IT	72000
6	Michael	Finance	80 000

8 SELECT Employee ID, First Name, Last Name, Department, Salary
FROM Employees
WHERE Department IN ('IT', 'Finance');

Employee ID	Name	Department	Salary
1	John	IT	70000
3	Bob	Finance	75 000
4	David	IT	72 000
6	Michael	Finance	80 000

9 SELECT Project ID, Project Name, Budget, status
FROM Projects
WHERE Status ! = 'Completed';

Project ID	Project Name	Budget	Status
102	Employee training	50 000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90 000	Ongoing
106	Risk Management	80000	Pending

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10 SELECT Project ID, Project Name, Budget, Status
FROM Projects
WHERE Budget > 70000 AND Status != 'Completed';

Project ID	Project Name	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Ongoing

11 SELECT Employee ID; First Name, Last Name, Department, salary, city
FROM Employees
WHERE City IN ('New York', 'Toronto')
ORDER BY salary DESC;

Employee ID	Name	Salary	City
6	Michael	80 000	New York
3	Bob	75 000	Toronto
1	John	70 000	New York

12 SELECT Employee ID, First Name, Last Name, Department, salary
FROM Employees
ORDER BY salary DESC
LIMIT 3;

Employee ID	Name	Salary
6	Michael	80 000
3	Bob	75000
4	David	72000

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13
```
SELECT Department,
Sum(salary) As totalsalary
FROM Employees
GROUP BY Department
ORDER BY Totalsalary DESC;
```

Department	Totalsalary
Finance	115 000
IT	142 000
Sales	65 000
HR	60 000

14
```
SELECT City,
AVG (salary) As Average_salary
FROM Employees
GROUP BY City
HAVING AVG (Salary) > 65 000;
```

City	Average_salary
New York	75 000
Toronto	75 000
London	72 000

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15 SELECT Department,
COUNT (Employee ID) AS Employee Count
FROM Employees
GROUP BY Department
HAVING COUNT (Employee ID) > 1 ;

Department	Employee_Count
IT	2
Finance	2

16 SELECT Status
COUNT (Project ID) AS Project Count
FROM projects
GROUP BY Status
HAVING COUNT (Project ID) >= 2;

Status	Project Count
Ongoing	2
Pending	2

17 SELECT e. EmployeeID, e. FirstName, e. LastName
Sum(p. Budget) AS total Project Budget
FROM Employees e
JOIN Projects p ON e. EmployeeID = p. EmployeeID
GROUP BY e. EmployeeID, e. FirstName, e. LastName
HAVING sum (p. Budget) > 150 000 ;

Employee ID	Name	Total project Budget
1	John	175 000

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